

ShopEZ

Full-Stack E-Commerce Web Application

Technical Documentation

Node.js • Express.js • MongoDB • HTML/CSS/JavaScript

1. Introduction

Project Title: ShopEZ – Full-Stack E-Commerce Platform

Developer: [Your Name]

Role: Full Stack Developer

2. Project Overview

Purpose: ShopEZ is a full-stack e-commerce web application that lets users browse products, register an account, log in securely, add items to a cart, and place orders through a simple, responsive interface. A protected admin dashboard allows product and order management.

Features:

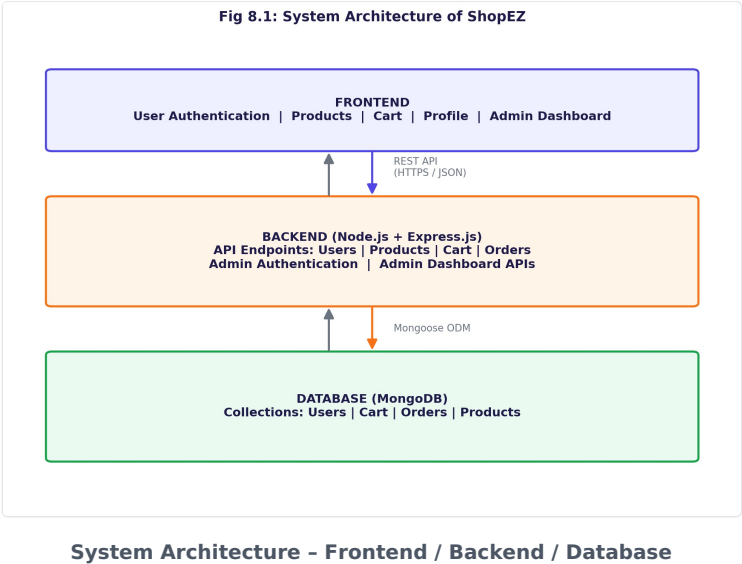
- User Registration and Login (JWT-based)
- Product Catalogue Browsing
- Shopping Cart Management
- Checkout with Address & Payment Method
- Order History via Profile Section
- Admin Login & Dashboard
- Product Management (Add / Update / Delete)
- MongoDB Database Integration

3. Architecture

Frontend: Built with plain HTML5, CSS3 and vanilla JavaScript – no framework or build step required.

Backend: Node.js with the Express.js framework, exposing REST APIs consumed by the frontend.

Database: MongoDB, accessed through Mongoose, storing Users, Products, Cart and Orders collections.



4. Setup Instructions

Prerequisites:

- Node.js & npm
- MongoDB (local instance or MongoDB Atlas)
- Code editor (e.g. VS Code)

Installation Steps:

```
cd backend
npm install
# configure .env (MONGO_URI, JWT_SECRET, PORT)
node seed.js          # loads sample products + admin account
```

```
npm start # backend runs on http://localhost:5000
```

Then open frontend/index.html directly in a browser, or serve it statically:

```
cd frontend
npx serve .
```

5. Folder Structure

Backend	Frontend
models/ (User, Product, Cart, Order) routes/ (userRoutes, productRoutes, cartRoutes, orderRoutes, adminRoutes) middleware/ (auth, errorHandler) index.js, seed.js, package.json, .env	css/style.css js/api.js, nav.js index.html, login.html, register.html cart.html, order.html, profile.html, admin.html

6. Running the Application

```
Backend:  cd backend && npm start  → http://localhost:5000
Frontend: open frontend/index.html (or npx serve frontend)
```

7. API Documentation

Module	Endpoints
User / Auth	POST /api/users/register · POST /api/users/login · GET/PUT /api/users/profile
Products	GET /api/products · GET /api/products/:id
Cart	GET /api/cart · POST /api/cart/add · POST /api/cart/remove
Orders	POST /api/orders · GET /api/orders · GET /api/orders/:id
Admin	POST /api/admin/login · POST/PUT/DELETE /api/admin/products · GET/PUT /api/admin/orders · GET /api/admin/users

8. Authentication

Authentication is implemented using JSON Web Tokens (JWT), with passwords hashed via bcrypt.js before storage.

- User Registration & Login
- Token Generation on successful login
- Protected Routes via an Express `protect` middleware
- Role-based Access Control via an `adminOnly` middleware

9. User Interface

- Home / Product Listing Page
- Registration Page
- Login Page
- Shopping Cart Page
- Order / Checkout Page
- Profile & Order History Page
- Admin Login & Dashboard

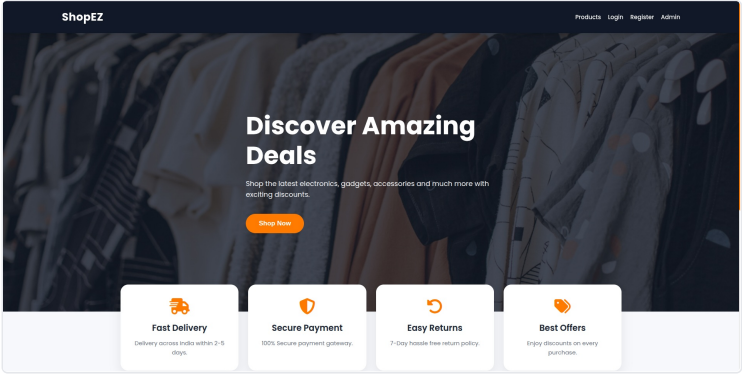
10. Testing

Testing performed manually across the following flows:

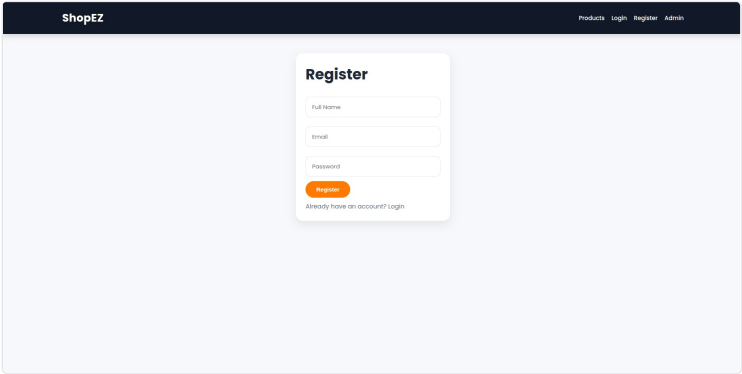
- User Registration & Login Testing
- Product Listing Fetch Testing
- Cart Add / Remove Testing

- Checkout & Order Creation Testing
- Profile Order History Testing
- Admin Login & Product Management Testing
- API Endpoint Testing (via Postman)

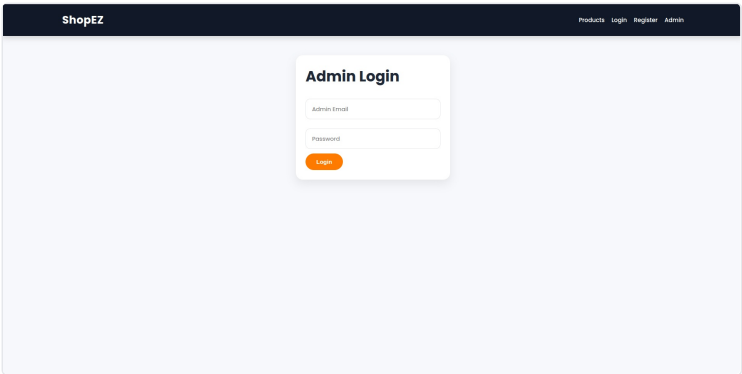
11. Screenshots / Output



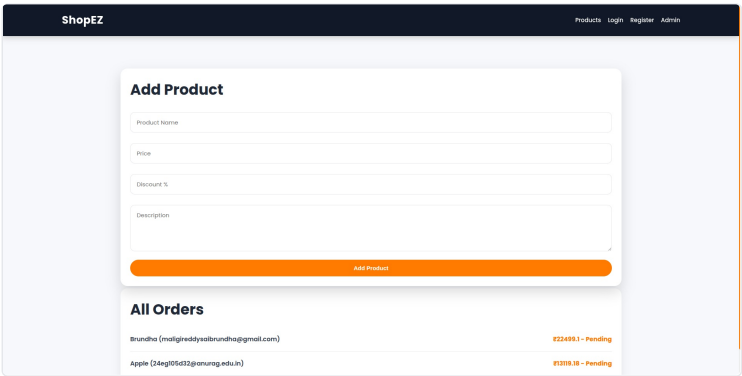
Home Page - Product Landing Page



User Registration Page



Admin Login Page



Admin Dashboard - Add Product & Manage Orders

ShopEZ

[Products](#)[Cart](#)[Profile](#)[Logout](#)[Admin](#)

My Profile

Name: Japple

Email: 24q9f5d3@parung.edu.in

Address: Not set

My Orders

Kindle Paperwhite xl

\$1299.00 - Pending

User Profile & Order History

12. Known Issues

- Payment processing is simulated; no live payment gateway is integrated yet.
- Single-vendor only – no multi-seller marketplace support.
- No product search or category filtering on the current frontend.
- No email/SMS notifications for order status updates.

13. Future Enhancements

- Live Payment Gateway Integration
- Product Search & Category Filtering
- Product Reviews & Ratings
- Order Tracking / Status Notifications
- Pagination for the Product Catalogue
- Admin Sales Analytics Dashboard
- Multiple Product Images per Listing
- Mobile Application Version